

Soumya Sri Kesani

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Education

University of Oklahoma – M.S. Data Science and Analytics (*GPA: 3.90/4.00*)

Aug 2024 – May 2026

CVR College of Engineering – B.Tech Computer Science Engineering (*GPA: 3.07/4.00*)

Graduated May 2024

Experience

Graduate Research Practicum

Jan 2026 – May 2026

Data Institute for Societal Challenges (DISC), University of Oklahoma

Norman, OK

- Analyzed **1.85M NASA VIIRS SNPP records** (2013–2024) across 53 African countries using R; applied coordinate reprojection, spatial clipping, deduplication, and missing-value validation to produce a clean, analysis-ready dataset from multi-year CSV archives.
- Achieved **14.1% improvement** in burned area detection over the MODIS MCD64A1 global standard; applied Mann-Kendall trend testing across 12 protected areas to identify statistically significant encroachment patterns; mean **Kappa 0.588** across six validation sites. [Live Demo](#)

IT Support and Service Operations

May 2025 – May 2026

University of Oklahoma

Norman, OK

- Managed security groups, access controls, and telecom workflows for campus-wide IT service requests; improved ticket resolution efficiency through structured workflow documentation.

Research Assistant

Jan 2023 – May 2023

CVR College of Engineering

Hyderabad, India

- Co-authored and published research on Geographic Alert Systems in IJCERT. [View Paper](#)

Projects

Heart Disease Risk Stratification: EarlyMed Dataset | *Python, Pandas, Scikit-learn, SHAP, Excel*

- Processed a **70,000-row, 19-feature clinical dataset** covering biometrics, lifestyle factors, and comorbidity indicators; engineered features including WHO BMI risk bands and stress-sleep interaction terms; trained Random Forest and Logistic Regression classifiers with **AUC 0.91** and F1-score of 0.88.
- Applied **SHAP explainability** to identify blood pressure, cholesterol, and family history as top risk predictors; delivered an interactive **Excel pivot dashboard** for risk stratification by age band, gender, and lifestyle cluster to support non-technical clinical stakeholders.

NLP-Based Customer Churn Prediction Pipeline | *Python, Snowflake, Streamlit, TF-IDF, Scikit-learn* | [Live Demo](#)

- Analyzed **767,000+ CFPB consumer complaints**; applied **TF-IDF vectorization** and trained Logistic Regression, Naive Bayes, and Random Forest classifiers on combined structured and text features; evaluated using F1-score, precision, and recall across class-balanced splits.
- Designed an end-to-end data flow from raw CSV ingestion into **Snowflake** through SQL-based transformation to a **live Streamlit triage dashboard** for real-time churn risk visualization.

Stock Market Interactive Dashboard | *Tableau, Excel*

- Cleaned and transformed multi-year stock market datasets spanning approximately **2.5 million records**; built interactive **Tableau dashboards** with time-series trend lines, volume overlays, and sector performance indicators with drill-down filtering.

US Natural Gas Consumption Forecasting (1973–2025) | *Python, Pandas, ARIMA, ETS, Scikit-learn*

- Built an end-to-end **time-series analysis pipeline** on 627 months of EIA data: STL decomposition, ADF stationarity testing, and rolling-window model comparison across baseline, regression, ETS, and ARIMA specifications using MAE, RMSE, and MAPE.

Energy Consumption Forecasting: Steel Manufacturing | *Python, Scikit-learn, K-Means, Isolation Forest*

- Processed **35,040 industrial sensor records**; built regression forecasting models, applied **K-Means clustering** (silhouette-scored) to identify 3 operational regimes, and implemented **Isolation Forest anomaly detection** to flag equipment fault conditions.

Technical Skills

Languages: Python (NumPy, Pandas, Scikit-learn, Matplotlib, SHAP), R (sf, terra, igrph, gstat, ggplot2), SQL

Analytics: Statistical Modeling, Predictive Analytics, Time-Series Forecasting, EDA, Feature Engineering, SHAP Explainability, A/B Testing, NLP Text Classification

Databases: MySQL, Snowflake, Microsoft Azure SQL, SQLite

Cloud / Tools: Microsoft Azure, Snowflake, Git, GitHub, Streamlit, Google Earth Engine

Visualization: Tableau, Power BI (DAX), Excel (Pivot Tables, Dashboards), Matplotlib, ggplot2

ML / Stats: Random Forest, Logistic Regression, Naive Bayes, ARIMA, ETS, K-Means, Isolation Forest, TF-IDF, Mann-Kendall Testing, STL Decomposition

Summary

Data analytics graduate with hands-on experience transforming large, complex datasets into actionable insights and stakeholder-ready visualizations. Proficient in Python, R, SQL, Tableau, Power BI, and Excel. Passionate about statistical modeling, predictive analytics, and end-to-end analysis workflows across clinical, environmental, and industrial domains. Committed to leveraging AI-augmented tools to surface insights faster and deliver measurable results.